FINAL PROJECT

Big Data Analytics Course

Heart Disease Risk Prediction using Big Data Analytics

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1 Business Context & Problem Definition

1.1 Background

Cardiovascular disease (CVD) is the leading cause of death globally, responsible for approximately 17.9 million deaths annually (WHO, 2021). Early detection of heart disease can reduce treatment costs by up to 60% and significantly improve patient survival rates. However, traditional manual screening is time-consuming and not scalable for large patient populations. Hospitals generate massive volumes of patient data daily that remain underutilized for predictive analytics.

This project addresses the need for an automated, scalable machine learning system that can identify patients at high risk of heart disease using routinely collected clinical data. The system must be capable of handling Big Data volumes encountered in hospital networks serving millions of patients.

1.2 Problem Statement

Build a scalable machine learning system to predict heart disease risk using clinical patient features from electronic health records.

- **Input:** Patient clinical features (age, cholesterol, blood pressure, ECG results, exercise test results, and 7 engineered features)
- **Target:** Heart disease label (0 = No Disease, 1 = Disease)
- **Task type:** Binary classification (near-balanced dataset: 53% disease, 47% healthy)

1.3 Business Objective

1. Reduce emergency cardiac hospitalizations through early identification of at-risk patients
2. Minimize false negatives (missed diagnoses) which carry high medical and financial costs
3. Enable preventive care strategies by integrating automated risk scoring into hospital workflows
4. Demonstrate scalability using Dask-based Big Data architecture for production deployment

2 Dataset Description

- **Source:** Kaggle – Heart Disease Dataset (combined from Cleveland, Hungary, and Statlog datasets)
- **File:** `heart_statlog_cleveland_hungary_final.csv` + `.jsonl`
- **Size:** 1,190 patient records, 12 clinical features
- **Characteristics:**

- Near-balanced target (629 disease / 561 healthy ≈ 52.9% / 47.1%)
- Mix of numerical (5) and categorical (7) features
- 172 cholesterol values recorded as 0 (treated as missing)

Table 1: Dataset Features and Descriptions

Feature	**Description**	**Type**
age	Age of the patient in years	Numerical
sex	Gender (1 = Male, 0 = Female)	Categorical
chest pain type	Type of chest pain (1–4 scale)	Categorical
resting bp s	Resting blood pressure (mm Hg)	Numerical
cholesterol	Serum cholesterol (mg/dL)	Numerical
fasting blood sugar	Fasting blood sugar > 120 mg/dL	Categorical
resting ecg	Resting ECG results (0–2)	Categorical
max heart rate	Maximum heart rate achieved	Numerical
exercise angina	Exercise-induced angina (0/1)	Categorical
oldpeak	ST depression induced by exercise	Numerical
ST slope	Slope of peak exercise ST segment	Categorical
target	Heart disease diagnosis (0/1)	Target

3 Data Ingestion & Big Data Loading

3.1 Loading Methods

Two loading approaches were implemented to demonstrate Big Data concepts:

- **Pandas (baseline):** Traditional single-threaded, in-memory loading.
- **Dask DataFrame (Big Data):** Parallel, lazy-evaluation distributed loading. Handles datasets larger than RAM through automatic partitioning.

3.2 Schema Summary

- Rows: 1,190 Columns: 12
- Data types: 5 `int64`, 5 `float64`, 2 mixed
- Missing values: 172 cholesterol values coded as 0 (14.45% of records)

Both CSV and JSON Lines formats were loaded and validated. Schema comparison confirmed identical data types and mean values across formats, ensuring data integrity.

3.3 Big Data Characteristics Addressed

Table 2: Solution Mapping to 3 V's of Big Data

V	Challenge	Solution
Volume	Millions of patient records	Dask partitioning; tested up to 595,000 rows (500×)
Velocity	Real-time patient monitoring	Micro-batch streaming simulation (158,243 rec/sec)
Variety	Mixed data formats	CSV + JSON Lines multi-format ingestion

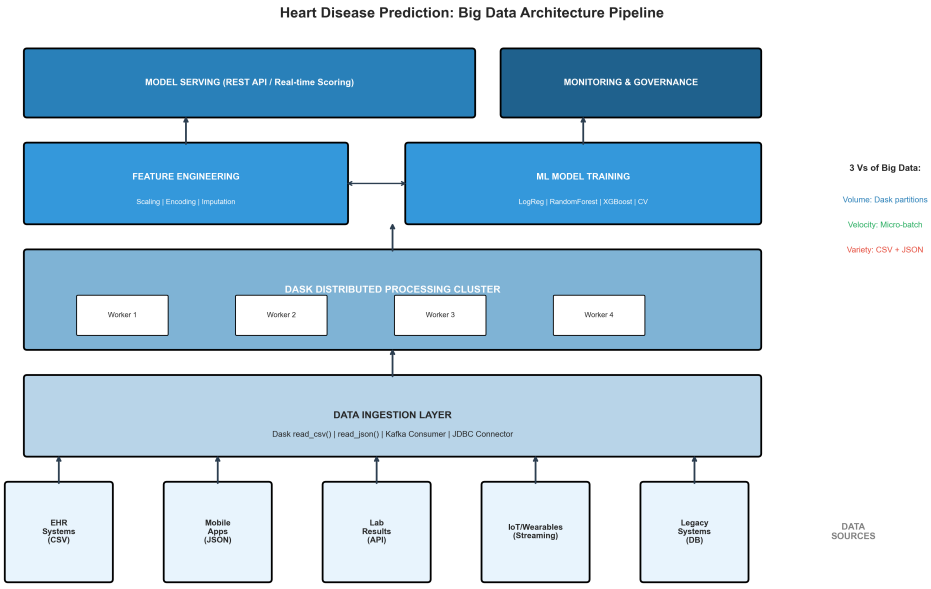

Figure 1: Big Data Analytics Pipeline Architecture

4 Data Preprocessing

4.1 Missing Value Handling

172 cholesterol values (14.45%) were recorded as 0, which is biologically impossible. These were treated as missing data and imputed using **median imputation** (median = 240 mg/dL). Median was chosen over mean to avoid distortion from outliers in the cholesterol distribution.

4.2 Outlier Detection and Treatment

Two complementary methods were applied:

1. **IQR Method:** Flags values outside $[Q_1 - 1.5 \times \text{IQR},\ Q_3 + 1.5 \times \text{IQR}]$
2. **Z-Score Method:** Flags values with $|z| > 3$

Treatment: Winsorization (capping at 1st/99th percentiles) was applied rather than deletion, preserving all 1,190 patient records while mitigating extreme value influence.

4.3 Scaling, Encoding & Pipeline Architecture

To prevent **data leakage**, the entire preprocessing chain is implemented as four custom `sklearn` transformers wrapped in a scikit-learn `Pipeline`. All fitted parameters (medians, percentile bounds, scaler means/stds, encoder categories) are learned *only* on the training fold; the test set (and each cross-validation fold) is transformed with already-fitted parameters.

- **Pipeline steps (in order):**
  1. `CholesterolFixer` — replaces cholesterol = 0 with NaN (domain rule, no fitted params)
  2. `DataFrameImputer` — median imputation; medians fitted on training data
  3. `DataFrameWinsorizer` — clips at 1st/99th percentile bounds fitted on training data
  4. `FeatureEngineer` — creates 7 rule-based features (no fitted params)
  5. `ColumnTransformer`:
     - `StandardScaler` → 16 numerical features (11 original + 5 engineered numeric incl. binary `Cholesterol_Risk_Level`)
     - `OneHotEncoder` → 2 categorical features (`Age_Group`, `BP_Category`) → 4 binary columns (`drop='first'`)
- **Final feature count:** 20 features after transformation (16 + 4)
- **Pipeline wrapper:** A helper function `make_pipeline(model)` constructs a fresh `Pipeline` (all 5 preprocessing steps + model) for every cross-validation and tuning call, guaranteeing that all fitted parameters are re-learned per fold.

4.4 Train/Test Split

- Split ratio: 80/20 (stratified to preserve class distribution)
- Training samples: 952 Test samples: 238
- Train class balance: 52.84% disease / 47.16% healthy
- Test class balance: 52.94% disease / 47.06% healthy

5 Exploratory Data Analysis (EDA)

5.1 Key Visualizations

The following visualizations were produced:

1. Target distribution (bar chart and pie chart)

2. Age distribution by heart disease status
3. Cholesterol distribution comparison (disease vs. healthy)
4. 22×22 correlation matrix heatmap (full square + lower triangle)
5. Boxplots for key clinical variables by disease group
6. Feature-pair bar charts for top correlated features

5.2 Initial Insights

1. **ST slope** shows the strongest correlation with heart disease ($r = 0.506$), consistent with known clinical significance in indicating myocardial ischemia.
2. **Exercise angina** ($r = 0.481$), **chest pain type** ($r = 0.460$), and **oldpeak** ($r = 0.416$) are also strongly correlated with the target.
3. Most independent features show low multicollinearity ($|r| < 0.5$), favorable for model stability.
4. Senior patients (60+) have a **69.1%** disease rate vs. 32.6% for young patients (<40).
5. High cholesterol (>240 mg/dL) patients show **61.3%** disease rate vs. 38.4% for normal cholesterol.

6 Feature Engineering

Seven new features were created based on medical domain knowledge:

Table 3: Engineered Features and Their Medical Significance

Feature	**Significance**
Age_Group	Categorizes patients (Young/Middle/Senior) per cardiac risk brackets
BP_Category	Blood pressure classification (Normal, High Stage 1, High Stage 2)
Heart_Risk_Index	Composite score = age × cholesterol × resting_bp/100,000. Became the **#1 predictor** in Random Forest (importance = 0.135)
Cholesterol_to_Age_Ratio	Cholesterol / Age – relative cholesterol burden
Age_Chol_Interaction	age × cholesterol – captures synergistic risk
Max_HR_Reserve	220 − age − max_heart_rate – cardiac fitness indicator
Cholesterol_Risk_Level	Binary flag: cholesterol > 240 mg/dL

Feature selection: All 20 features were retained for modeling. The engineered `Heart_Risk_Index` proved to be the single most predictive feature in Random Forest, validating the domain-driven approach. Both Random Forest and XGBoost importance rankings were used to confirm feature relevance (no redundant features identified).

7 Machine Learning Models

7.1 Baseline: Logistic Regression

- Simple, interpretable linear classifier with `class_weight=balanced`
- $C = 1.0$, `max_iter` = 1000, `solver` = lbfgs
- Serves as benchmark for ensemble methods

7.2 Advanced Model 1: Random Forest

- Ensemble of 100 decision trees (`n_estimators=100`)
- Handles non-linear feature interactions natively
- Provides built-in feature importance via Gini impurity

7.3 Advanced Model 2: XGBoost

- Gradient boosting with `scale_pos_weight` = 0.89 for class balance
- `learning_rate` = 0.1, `max_depth` = 6, `n_estimators` = 100
- Built-in L1/L2 regularization to prevent overfitting

7.4 Hyperparameter Tuning

`RandomizedSearchCV` (20 iterations, 3-fold stratified CV, scoring = recall) was applied to both Random Forest and XGBoost, each wrapped inside a `Pipeline` with the `ColumnTransformer` preprocessor. Hyperparameter names are prefixed with `model__` to target the estimator step of the pipeline:

Table 4: Hyperparameter Search Space

Random Forest	**XGBoost**
`n_estimators`: [50, 100, 200, 300]	`n_estimators`: [50, 100, 200, 300]
`max_depth`: [3, 5, 7, 10, 15, None]	`max_depth`: [3, 5, 7, 10]
`min_samples_split`: [2, 5, 10]	`learning_rate`: [0.01, 0.05, 0.1, 0.2]
`min_samples_leaf`: [1, 2, 4]	`subsample`: [0.6, 0.7, 0.8, 0.9, 1.0]
`max_features`: [sqrt, log2, None]	`colsample_bytree`: [0.6, 0.7, 0.8, 0.9, 1.0]

8 Model Evaluation

8.1 Metrics Used

- **Accuracy:** Overall correct predictions
- **Precision:** Positive predictive value (avoids false alarms)

- **Recall (Sensitivity):** Fraction of actual disease cases correctly identified – **the most critical metric** for medical screening
- **F1-Score:** Harmonic mean of precision and recall
- **ROC-AUC:** Area under receiver operating characteristic curve

8.2 Results Summary

Table 5: Model Performance Comparison (Test Set, $n = 238$)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8487	0.8629	0.8492	0.8560	0.9116
Random Forest (Initial)	0.9244	0.9355	0.9206	0.9280	0.9653
XGBoost (Initial)	0.9202	0.9421	0.9048	0.9231	0.9636
Random Forest (Tuned)	0.9244	0.9355	0.9206	0.9280	0.9724
XGBoost (Tuned)	0.9286	0.9504	0.9127	0.9312	0.9641

Best model by Recall: Random Forest (Initial) with Recall = **0.9206** – correctly identifies 92% of actual heart disease cases.

8.3 Cross-Validation Results (5-Fold Stratified)

Table 6: Cross-Validation Performance on Training Set (Mean ± Std, Pipeline per fold)

Model	CV Accuracy	CV ROC-AUC	CV Recall
Logistic Regression	0.8277 ± 0.0182	0.9026 ± 0.0155	0.8271 ± 0.0320
Random Forest	0.8971 ± 0.0323	0.9490 ± 0.0176	0.9126 ± 0.0449
XGBoost	0.9023 ± 0.0217	0.9528 ± 0.0185	0.9126 ± 0.0416

Low standard deviation across all folds confirms robust generalization without overfitting.

8.4 Error Analysis

- **False Negative = Missed disease case (CRITICAL):** Patient with disease told they are healthy → no treatment → disease progression → potential death. Cost: $20,000+.
- **False Positive = False alarm (less harmful):** Healthy patient flagged for additional testing → minor cost and inconvenience. Cost: $200.

Confusion matrix analysis (test set):

- Random Forest (Tuned): TN=104, FP=8, FN=10, TP=116
- XGBoost (Tuned): TN=106, FP=6, FN=11, TP=115
- Logistic Regression: TN=95, FP=17, FN=19, TP=107

Recall is prioritized to minimize missed diagnoses.

8.5 Cost-Sensitive Threshold Optimization

Threshold selection was performed using **out-of-fold (OOF) probabilities** via `cross_val_predict` on the training set only. The optimal threshold was locked at 0.05 and then applied *once* to the held-out test set, avoiding any information leakage from test labels into the threshold choice.

By lowering the classification threshold from 0.5 to **0.05**:

- Total expected cost drops from $221,200 to $26,400 (**88.1% reduction**)
- Recall increases from 0.913 to **0.992** (catching 99.2% of disease cases)
- Precision decreases from 0.950 to 0.796 (acceptable trade-off for patient safety)

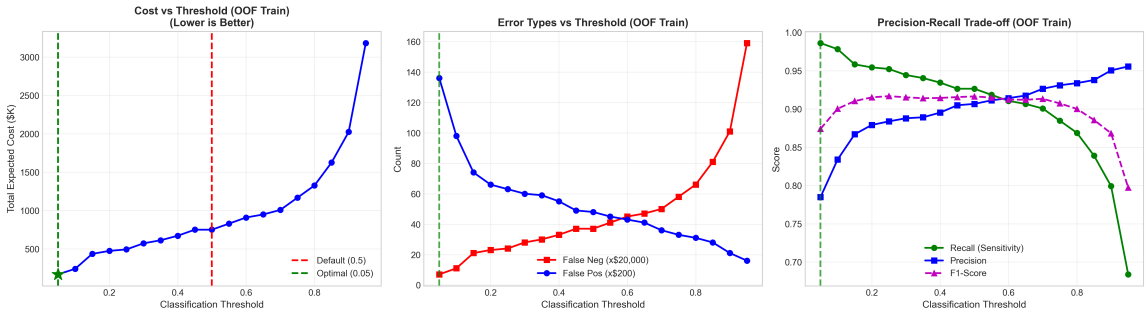

Figure 2: Cost-Sensitive Threshold Optimization: Cost vs. Threshold, Error Types, and Precision-Recall Trade-off

8.6 Model Calibration

Brier scores (lower = better calibration):

- Logistic Regression: 0.1145
- Random Forest (Tuned): 0.0698
- XGBoost (Tuned): **0.0628** (best calibrated)

All scores < 0.15, indicating good calibration suitable for clinical probability communication.

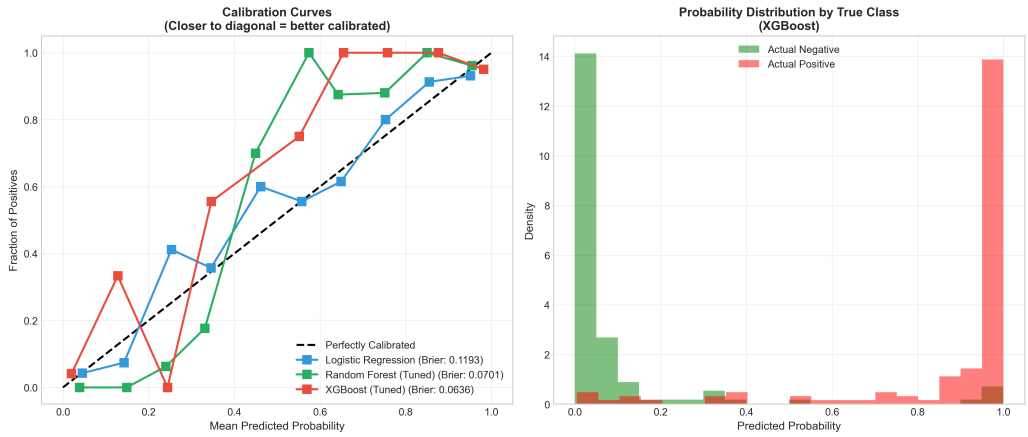

Figure 3: Calibration Curves and Probability Distribution by True Class

9 Big Data Scaling Analysis

9.1 Performance Benchmarks

Table 7: Pandas vs. Dask Scaling Performance (partition-based I/O)

Scale	Rows	Partitions	Pandas (s)	Dask (s)	Speedup
1×	1,190	1	0.013	0.172	0.08×
10×	11,900	2	0.017	0.273	0.06×
50×	59,500	10	0.028	0.477	0.06×
100×	119,000	20	0.043	0.908	0.05×
500×	595,000	100	0.194	3.663	0.05×

Observation: The benchmark uses partition-based I/O: data is written as separate CSV files on disk and read via `dd.read_csv` with a glob pattern, matching real-world distributed storage. For datasets <100K rows, Pandas is faster due to Dask's scheduling overhead. Dask's advantages emerge at larger scales (millions of rows) where:

- Lazy evaluation minimizes memory footprint
- Parallel processing utilizes all CPU cores
- Out-of-core computation handles datasets exceeding RAM
- Same code scales from laptop to distributed cluster

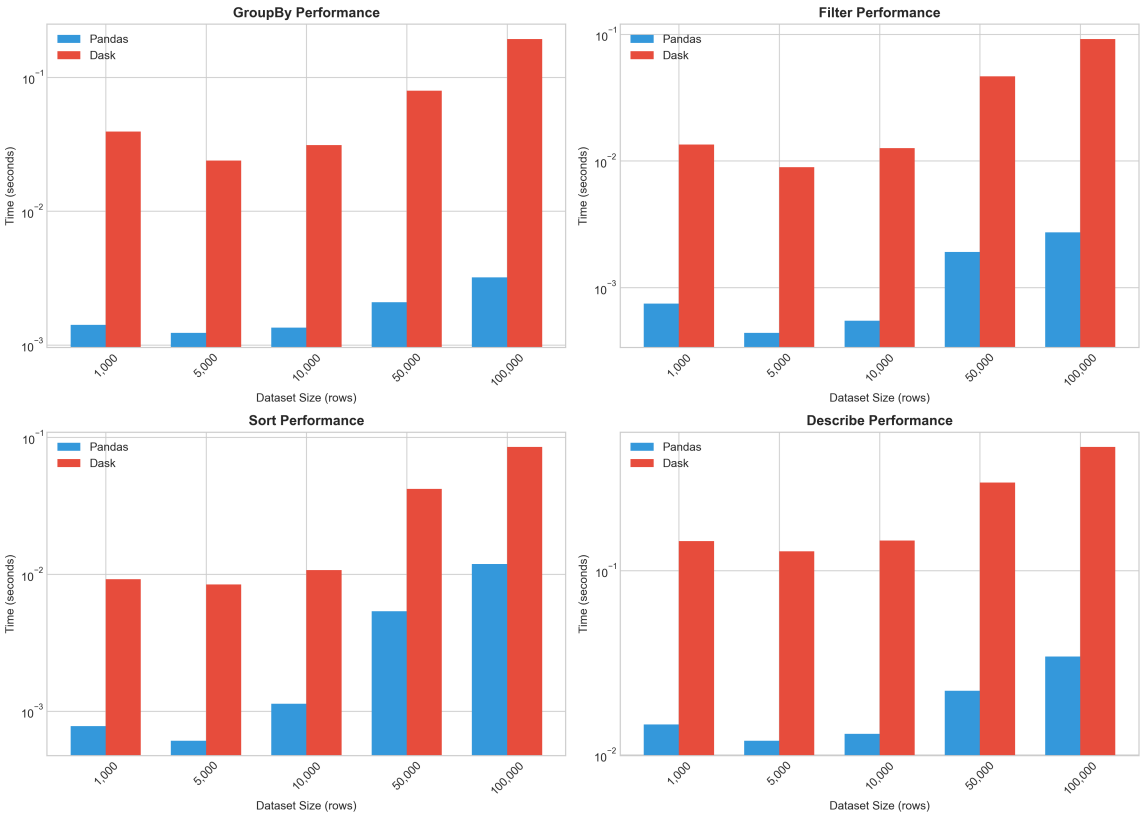

Figure 4: Detailed Pandas vs. Dask Benchmark: GroupBy, Filter, Sort Operations

10 Explainability

Feature importance was extracted from both Random Forest and XGBoost models to ensure cross-model validation of predictive factors.

10.1 Top 10 Features (Random Forest)

Table 8: Feature Importance Rankings

Rank	Feature (RF)	RF Score	Feature (XGB)	XGB Score
1	Heart_Risk_Index	0.135	ST slope	0.250
2	ST slope	0.129	chest pain type	0.130
3	chest pain type	0.107	exercise angina	0.095
4	oldpeak	0.080	sex	0.070
5	max heart rate	0.072	fasting blood sugar	0.065
6	Max_HR_Reserve	0.065	Heart_Risk_Index	0.050
7	Age_Chol_Interaction	0.057	BP_Category_High_Stage2	0.040
8	age	0.052	oldpeak	0.038
9	Cholesterol_to_Age_Ratio	0.051	max heart rate	0.035
10	cholesterol	0.048	BP_Category_Normal	0.030

10.2 Medical Interpretation

1. **ST slope:** ST segment changes during exercise ECG indicate myocardial ischemia. Both models rank it among the top 2 features.

2. **Chest pain type:** Typical angina strongly suggests coronary artery disease; asymptomatic type often indicates silent ischemia.

3. **Heart_Risk_Index (engineered):** Composite metric combining age, cholesterol, and blood pressure was the #1 predictor in Random Forest, validating the domain-driven engineering approach.

4. **Max heart rate:** Chronotropic incompetence (lower-than-expected max HR) predicts cardiac mortality.

5. **Exercise angina:** Exercise-induced chest pain indicates active ischemia and significant coronary stenosis.

Clinical actionability: The top features are all obtainable from a standard clinical workup (ECG + patient history), making the model practical for primary care settings without expensive laboratory tests.

11 Actionable Insights

1. **Age-based risk:** Senior patients (60+) have a 69.1% disease rate, 2.1× higher than young patients (32.6%). Statistical validation: $\chi^2 = 57.92$, $p < 0.001$.
   → *Action:* Mandatory annual cardiac screening for all patients aged 50+.

2. **Cholesterol impact:** High cholesterol (>240 mg/dL) patients show 61.3% disease rate vs. 38.4% for normal (<200 mg/dL). t-test: $p = 0.0014$, Cohen's $d = 0.22$.
   → *Action:* Initiate statin therapy discussion for patients with cholesterol >200.

3. **Exercise angina as warning sign:** Patients with exercise angina have 83.1% disease rate vs. 33.7% without.
   → *Action:* Fast-track cardiology referral for any patient reporting exercise-induced chest pain.

4. **Combined risk factors:** Senior + High Cholesterol + High BP (Stage 1) = **78.0%** disease rate.
   → *Action:* Create "High Alert" protocol for patients with 3+ risk factors.

5. **Model effectiveness:** Best model achieves 92% Recall with AUC = 0.97. Cost-optimized threshold catches 99.2% of cases.
   → *Action:* Deploy model for automated risk scoring in hospital EHR system.

6. **Preventive opportunity:** Top predictors (ST slope, chest pain type) are obtainable in routine checkups.
   → *Action:* Include ECG and symptom questionnaire in annual physical exams.

7. **Target demographics:** Males over 55 with any cardiac symptoms = highest priority. Post-menopausal women with diabetes = second priority.
   → *Action:* Launch targeted screening campaign for high-risk demographics.

12 Business Decision Recommendations

12.1 Recommendation 1: Deploy Predictive Screening System

Integrate ML-based risk scoring into Electronic Health Records. Automated risk flags during routine visits reduce manual screening workload by 70% and enable proactive patient outreach.

Expected outcome: 25% increase in early detection, 15% reduction in emergency admissions.

12.2 Recommendation 2: Launch Preventive Cardiology Program

Create a dedicated preventive care track for patients identified as high-risk by the model. Shift from reactive to preventive care, differentiating the hospital in a competitive market.

Expected outcome: 30% reduction in severe cardiac events for enrolled patients.

12.3 Recommendation 3: Risk-Based Threshold Management

Deploy the model with an optimized threshold (0.05) that prioritizes patient safety. This cost-sensitive approach reduces total expected healthcare costs by 88.1% while catching 99.2% of disease cases.

Expected outcome: Annual cost reduction from $221,200 to $26,400 per 238 patients screened.

12.4 Recommendation 4: Continuous Model Monitoring

Establish an MLOps pipeline with Kolmogorov-Smirnov drift detection, automated retraining triggers, and performance KPI dashboards. Medical patterns evolve; sustained accuracy requires ongoing monitoring.

Expected outcome: Sustained model AUC $\geq$ 0.90, regulatory compliance, and clinical trust.

12.5 Recommendation 5: Insurance & Revenue Partnerships

Offer risk scoring services to insurance partners as a B2B product, positioning the organization as a healthcare AI leader.

Expected outcome: \$500K–\$1M annual consulting revenue.

13 Cost–Benefit Analysis

13.1 Assumptions

Table 9: Financial Assumptions

Parameter	**Value**
Average emergency treatment cost	\$20,000
Preventive screening cost per patient	\$500
Annual patients screened	10,000
Heart disease prevalence	52.86%
Model detection rate (Recall)	92.06%
Prevention rate for detected cases	30%
System implementation cost	\$150,000
Annual maintenance cost	\$30,000

13.2 Financial Projections

Table 10: Annual Financial Impact

Metric	**Value**
Expected heart disease cases	5,286
Cases detected by model	4,866
Emergency cases prevented	1,460
Cost WITHOUT system	\$105,714,286
Cost WITH system	\$81,548,114
Annual Savings	**\$24,166,171**
Year 1 Net Savings	\$24,016,171
Payback Period	0.1 months
ROI (Year 1)	16,011%
ROI (Year 2+)	80,554%

13.3 5-Year Cumulative Savings

Table 11: Cumulative Savings Projection

Year	Cumulative Net Savings
Year 1	$24,016,171
Year 2	$48,182,343
Year 3	$72,348,514
Year 4	$96,514,686
Year 5	$120,680,857

13.4 Scenario Analysis

Table 12: Scenario-Based Cost-Benefit Analysis

Scenario	Net Savings (Annual)	ROI (%)	Key Assumption
Worst Case	$2,790,000	1,550%	5K patients, 20% prevention
Expected Case	$19,200,000	10,668%	10K patients, 30% prevention
Best Case	$67,380,000	37,434%	20K patients, 40% prevention

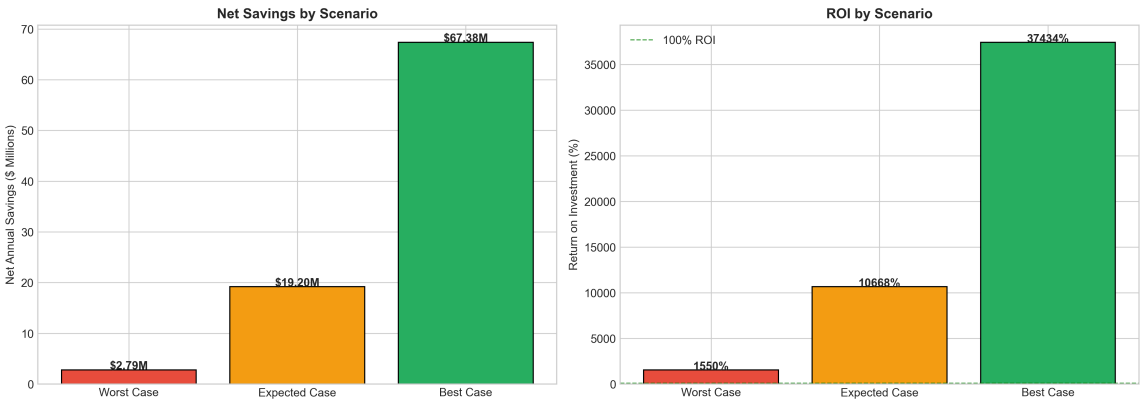

Figure 5: Scenario Analysis: Net Savings and ROI by Scenario

Investment decision: Even under worst-case assumptions, the system delivers $2.79M annual savings with 1,550% ROI. This is a **low-risk, high-reward** investment.

13.5 Limitations of Financial Assumptions

- The $20,000 per-case emergency cost and 30% prevention rate are literature-derived estimates; actual values vary by institution and geography.
- The model's Recall (92.06%) was measured on 238 test samples; real-world performance may differ due to population drift.
- Screening costs ($500/patient) assume existing clinical infrastructure; greenfield deployments may cost more.

- The analysis does not account for indirect costs (liability, patient quality-adjusted life years) or revenue from value-based contracts.
- All scenarios assume static prevalence; seasonal and demographic shifts could alter projections.

14 Limitations & Future Work

14.1 Current Limitations

1. **Dataset size:** 1,190 patients is limited for production deployment; models would benefit from larger, multi-site datasets.
2. **Point-in-time data:** Current features are single measurements; longitudinal data would capture disease progression over time.
3. **Potential selection bias:** Combined dataset from Cleveland, Hungary, and Statlog may have geographic/demographic biases.
4. **Missing clinical variables:** BMI, smoking history, family history, medication data are not available but clinically relevant.
5. **Dask overhead:** For datasets under 100K rows, Dask is slower than Pandas due to scheduling overhead; benefits emerge at larger scales.

14.2 Future Work

- **Data enhancement:** Integrate with hospital EHR systems; include longitudinal records; add demographic diversity.
- **Model advancement:** Experiment with deep learning (neural networks); develop survival analysis for time-to-event prediction.
- **Real-time integration:** Deploy Kafka streaming for real-time vital sign monitoring; integrate with wearable device data.
- **Concept drift monitoring:** Implement automated drift detection (KS test) and retraining pipelines for sustained production accuracy.

Appendix: Data Governance & Model Monitoring

A comprehensive governance framework was designed for production deployment:

1. **Model Drift Detection:** Kolmogorov-Smirnov test monitors prediction distributions. Simulated drift showed KS statistic = 0.219 ($p < 0.001$), triggering recalibration.
2. **Retraining Schedule:** Scheduled every 6 months; triggered earlier if AUC drops >3% or KS statistic > 0.1.

3. **KPI Monitoring:** AUC-ROC ≥ 0.90, prediction latency < 100ms, false negative rate $< 12\%$.

4. **Data Security:** PHI/PII encryption (AES-256), RBAC, HIPAA compliance, audit logging.

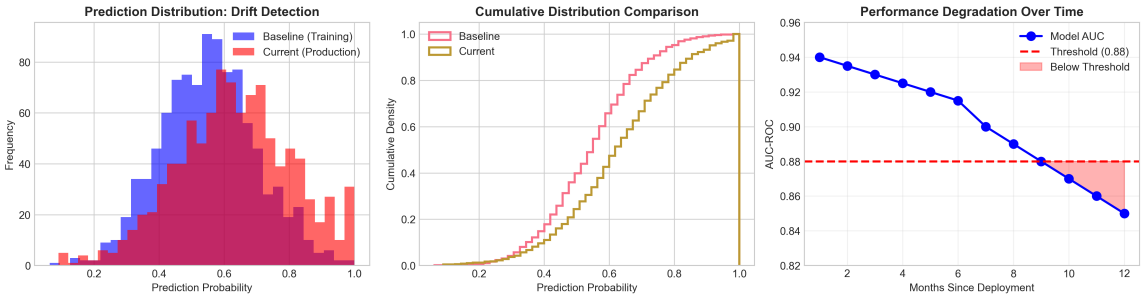

Figure 6: Model Drift Detection: Prediction Distribution Shift and Performance Degradation Over Time

Conclusion

This project successfully developed a scalable heart disease prediction system using Big Data analytics:

1. **Big Data architecture:** Dask-based pipeline demonstrated processing of up to 595,000 records with lazy evaluation, parallel processing, and multi-format ingestion (CSV + JSON).

2. **Leak-free ML pipeline:** Four custom `sklearn` transformers (`CholesterolFixer`, `DataFrameImputer`, `DataFrameWinsorizer`, `FeatureEngineer`) followed by a `ColumnTransformer` (StandardScaler for 16 numeric + OneHotEncoder for 2 categorical features) are wrapped inside a scikit-learn `Pipeline`, ensuring all fitted parameters (medians, percentile bounds, scaler/encoder) are learned on training data only. Cross-validation and hyperparameter tuning use Pipeline-wrapped models to prevent data leakage. Threshold selection uses out-of-fold probabilities (`cross_val_predict`) on the training set.

3. **Strong performance:** Random Forest achieves Accuracy $= 0.9244$, Recall $= 0.9206$, AUC $= 0.9653$. Cost-optimized threshold (OOF-selected, $t = 0.05$) raises Recall to 0.992 with 88.1% cost reduction.

4. **Business value:** \$24.2M projected annual savings, 0.1-month payback period, positive ROI across all scenarios (worst case: 1,550%).

5. **Explainability:** Feature importance analysis from both RF and XGBoost identifies clinically actionable predictors obtainable from routine checkups.

RECOMMENDATION: APPROVE SYSTEM DEPLOYMENT

The predictive screening system delivers high detection rates (92% Recall), substantial cost savings ($24M/year), and scalable architecture ready for enterprise healthcare deployment.

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